

Jianxiao(Shawn) Cai

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Education

University of California, San Diego

M.S. in Electrical Engineering (Computer Engineering), GPA: 3.65

La Jolla, CA

Sep 2024 – Mar 2026

Shanghai Normal University

B.Eng. in Electronic Information Engineering, GPA: 3.77

Shanghai, China

Sep 2020 – Jun 2024

Work Experience

Pixel Blue LLC

Founder & Software Engineer

San Diego, CA

Mar 2026 – Present

- Launched **Prec**, an iOS recurring-checklist app (SwiftUI + SwiftData), from concept to App Store submission, owning full product lifecycle: market research, PRD, UX design, **Swift** implementation, and release pipeline.
- Conducted competitive analysis across 5+ apps (PackPoint, Packr, TripList) to identify an underserved daily-reset use case; defined a 3-tier pricing model (Free / \$2.99/mo / \$6.99/mo) and brand positioning from scratch.
- Leading product discovery for a Family Care coordination app targeting 63M US caregivers, conducting community observation, and competitive benchmarking of existing tools to define MVP scope and distribution strategy.

ivector

Software Engineering Intern

Sacramento, CA

Jul 2025 – Sep 2025

- Designed 15+ UI screens in **Figma** and translated product requirements into user flows and interface prototypes.
- Implemented 3 core application pages (home, login, registration) in **Unity** using **C#**, building navigation logic and interactive UI components.
- Integrated **Firebase** Authentication supporting 4 flows (login, registration, password reset, email verification), establishing secure user account management within the app.

FORVIA HELLA

Advanced Engineering Intern

Shanghai, China

Jan 2024 – May 2024

- Developed BLDC motor control logic in **Simulink** with auto-generated code, reducing manual embedded coding by 40%.
- Established a reusable **Model-Based Design** workflow adopted across engineering projects, improving development consistency and iteration speed.

Projects

Tech Community Platform – Spring Boot, Redis, Elasticsearch, WebSocket, MySQL

Jan 2026 – Mar 2026

- Built a scalable full-stack community platform with 11 frontend pages and 30+ REST APIs across 9 modules (posts, comments, likes, follows, DMs, notifications, search, auth).
- Optimized read APIs via **Redis** feed caching, **Elasticsearch** query tuning, and payload trimming; reduced posts P95 latency from 6464ms to 631ms and search P95 from 5271ms to 462ms under 50-concurrent load.
- Implemented real-time messaging via **WebSocket** + Redis pub/sub, achieving sub-100ms message delivery and eliminating full-page reload flicker.
- Modeled social interaction data flows including follow graph traversal, notification fan-out, threaded comments, and unread-state tracking across user activity pipelines.
- Established CI/CD infrastructure with 3 **GitHub Actions** pipelines, 18 test classes, and 7 deployment scripts for automated validation and zero-downtime deployment.

shawnidea.com – WordPress, PHP, AWS EC2/S3, Cloudflare, GitHub Actions

Jan 2026 – Feb 2026

- Authored a custom **WordPress** theme from scratch with 5 page templates (homepage, projects, blog, resume, contact), implementing a consistent design system using **CSS** custom properties and responsive layout.
- Developed and iterated on the theme using an AI-assisted workflow (**GitHub Copilot** + **VSCode**), committing code changes directly to **GitHub** and iterating rapidly via AI-generated diffs.
- Deployed on **AWS EC2/S3** with **Cloudflare** DNS; fully automated CI/CD via **GitHub Actions** with 100+ releases.

Skills

- Languages:** Java, Python, TypeScript, JavaScript, Swift, C/C++, Go, C#, SQL
- Backend & Systems:** Spring Boot, Redis, Elasticsearch, WebSocket, MySQL, PostgreSQL, MyBatis, Flask
- Mobile:** SwiftUI, SwiftData, iOS Development, App Store deployment
- Tools & Infra:** Docker, AWS, GitHub Actions, Git, Linux, Figma